

Final Project details for MA 5104

Statistical Inference and Learning, July-Dec 2026

OVERVIEW

In the fields of financial markets, pandemic, sports, judiciary, and policy making, you may have noticed how the same data can be interpreted differently by different people. Critical thinkers need to be able to use the tools of probability and statistical analysis to meaningfully interpret data. We will use public media-based resource selected by you with our mutual agreement. I have given below a representative examples of podcasts. You could choose any other podcast/news articles/YouTube videos/opinion polls. The project will involve thinking about how we communicate data to the general public and the tools used for interpreting real-life data. The assessment will be based on a 3MT (“three-minute thesis”) presentation. See here for an example: <https://www.youtube.com/watch?v=zqb6Sq4Ml68> and here for tips on making it successful: https://www.youtube.com/watch?v=D_1MK3Ub9QM.

As discussed earlier, each student will complete a group project. Groups may have at most two members, and each group will analyze a different public media.

Every student should take on a distinct and meaningful role within the group, and these roles must be communicated to the instructor.

- If you choose to work alone, that is perfectly acceptable.
- If you choose to work in pairs, please discuss and agree on responsibilities early. This will help avoid misunderstandings or disagreements later on.
- In the rare case that serious conflicts arise within a group, please inform the instructor as soon as possible so that a fair resolution can be reached.

The project is worth **100 points**, divided into three components:

- **Initial Report (5 point):** due **September 16, 2026**.
- **Presentation (70 points):** Each group will give a 5-minute presentation on their project outcomes. These will be scheduled during the final lecture hours, or outside class hours if necessary. The 70 points consists of speaking skills, explanation of description and general problems, timekeeping, creativity.
- **Slide (25 points):** due immediately before presentation. These 25 points are essentially for the appearance of the one slide you make.

WEBLINKS

- (1) Google Drive link: https://docs.google.com/spreadsheets/d/10pnaTa2wNzogMLZ7moTpBpa_Lju7frU17iqQSczKREU/edit?usp=sharing
- (2) More or Less: Behind the Stats: <https://www.bbc.co.uk/programmes/p02nrss1/episodes/> Some interesting episodes from the podcast are: <https://www.bbc.com/audio/play/p0m0x53c>, <https://www.bbc.com/audio/play/p0lyglsw>, <https://www.bbc.com/audio/play/p0f820pq>, <https://www.bbc.com/news/health-57552527>.

This is a two-step assignment where you are required to complete the following tasks:

STEP 1: SELECTING THE MEDIA

Finalize a group mate. Based on the provided podcast link (number (2) above) or any other relevant public media available in the public domain, select an episode that meets the following criteria:

- The selected media is centered around applying statistics or probability in real life.
- The selected media has a duration of approximately 15 minutes (longer but < 30 min acceptable).
- When submitting the selected media, ensure that there is no clash in the selection of media among the participating students.

Once finalized, update the selected media (URL, title, and run-time) corresponding to your name in the Google Sheet (link no. 1 above) before September 16. The media will be approved by September 25 by the instructor.

PRESENTATION

Each team will give a **5-minute presentation** summarizing the outcomes of their project. Only one page/slide is allowed. You are expected to deliver a concise, single-slide, 5-minute oral presentation in 3MT (“three-minute thesis” inspired) style adapted for a 5-minute slot. The emphasis is on clarity, storytelling, and exact references to the public media (use timestamps when making claims about the media).

BASIC RULES (MUST-FOLLOW)

- Duration: **5 minutes** maximum. Practice with a timer and aim for 4:30–5:00.
- Slide: **One static slide only**. No animations, no multiple slides, no video playback from the slide.

- Props: No physical props or scripts visible to the audience. Use the one slide as the visual anchor.
- Language: Academic tone but conversational delivery. Do *not* read the slide verbatim.
- Paraphrase: Do not quote long sentences from the public media. Present findings and observations in your own words.
- Evidence linkage: Whenever you refer to data, methods, or claims presented in the media, explicitly state the **timestamp(s)** (e.g., “at 02:15 – 02:34”) during which the evidence appears.

MESSAGE OF THE PRESENTATION

The presentation must essentially:

- (1) Summarize the statistical tools employed for the subject matter discussed in the selected media. Identify and explain the statistical methodologies employed, their significance, and their relevance to the subject matter. Avoid directly quoting sentences from the public media. Instead, express your observations in your own words.
- (2) Identify and describe the different input data utilized in the selected media. Provide a comprehensive account of the sources of the data used, including relevant details such as data collection methods, data types, and data preprocessing techniques.
- (3) Summarize the findings derived from the statistical analysis conducted in the selected media. Present the key results and insights obtained, providing an interpretation of the outcomes. Ensure that your statements about the data, adopted statistical tools, or findings are appropriately linked to the relevant time stamps in the public media.
- (4) Discuss the potential limitations of the adopted methodology in the selected media. Analyze factors that could impact the accuracy, reliability, or generalizability of the statistical analysis. Provide a critical assessment of these limitations and propose possible ways to mitigate them or alternative approaches to address them.
- (5) Explore real-life scenarios, distinct from the selected media, where similar data extraction methodologies and statistical tools can be applied. Describe these scenarios and elucidate how the adoption of similar approaches could yield valuable insights or practical benefits. Support your discussion with relevant examples and highlight the specific time stamps in the public media that showcase the applicability of such methodologies.

SLIDE LAYOUT (SINGLE SLIDE)

Design the single slide to support, not replace, your spoken narrative. Suggested layout:

- (1) Title (top): 10–12 words maximum.
- (2) One-line context / hook (beneath title): 1 short sentence.
- (3) Central visual: a single figure, table excerpt, schematic, or flowchart summarising the method or the key result. Keep text on the figure minimal.
- (4) Right or bottom: 3 short bullets (**Methods**, **Key result**, **Takeaway**). Each bullet 8–12 words.
- (5) Footer (small font): citation to the media (title, outlet, date) and the key timestamp(s) you will reference.

RECOMMENDED 5-MINUTE STRUCTURE (TIMESTAMPS = SPEAKER TIMING)

- **0:00–0:40 — Hook and context (40 s)**: One engaging sentence to place the problem. Mention the media source and the most relevant timestamp for the context (e.g., "see 00:15–00:40").
- **0:40–1:30 — Data and inputs (50 s)**: Describe what data the media uses and where it came from (type, collection, preprocessing). When you mention each data source, speak the corresponding timestamp (e.g., "at 01:05 the presenter cites survey data").
- **1:30–2:40 — Statistical tools and approach (70 s)**: Summarize the statistical methods used, their purpose, and why they were appropriate. Link tools to timestamps where they are discussed or applied.
- **2:40–3:40 — Key findings (60 s)**: State the main results and their practical interpretation. Tie each result to the media timestamp where it appears.
- **3:40–4:20 — Limitations and critique (40 s)**: Briefly list 2–3 limitations of the adopted methodology and one concrete mitigation or alternative. Reference the moment in the media that suggests or omits those limitations.
- **4:20–5:00 — Broader applications and closing takeaway (40 s)**: Give 1–2 real-world scenarios where the same extraction methods and statistical tools would be useful. Point to the slide's footer timestamp(s) or a specific clip that demonstrates transferability. Finish with one crisp takeaway sentence.

SPEAKER TIPS

- Practice until your delivery fits the 5-minute structure naturally. Time each section during rehearsal.

- Use plain language for explanations; keep technical jargon minimal and define any essential term in 1–2 words.
- When you cite a timestamp, speak it clearly (“at two minutes and fifteen seconds, 02:15”). This ties your claim to evidence and satisfies assessment rules.
- Emphasize transitions between sections with short signposting phrases (“Now: data”, “Next: results”).
- Keep voice energy varied; pause briefly after important lines to let them land.
- Avoid reading: use the slide as a visual anchor and keep notes to 3–4 short prompts only.
- Please maintain professionalism and respect during presentations. Listen carefully to other teams and avoid unnecessary disruptions.
- The presentation must paraphrase — show that you can summarise the media’s content in your own words.
- Every factual claim about the media’s data, tools, or findings should be traceable to a timestamp; include those timestamps verbally and on the slide footnote.
- Maintain clarity, accurate referencing, and honest critique; if you adapt any numbers from the media, indicate the timestamp and any preprocessing assumptions you infer.